

Foundations of Artificial Intelligence - 2026.1

Sequential Decisions under Risk - Markov Decision Processes and Policies

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Decision Making Situations

Decision Situation Matrix

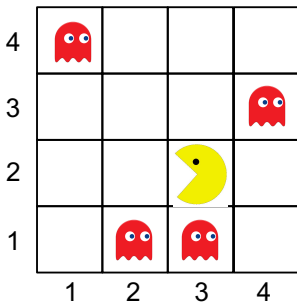
		Action Effects		
Horizon		Certainty	Risk	Uncertainty
	Atomic	✓	✓	
	Sequential	✓	NOW: 0%	

Course Outline

1. Markov Decision Processes
2. Trials
3. Policies and the Decision Problem

Markov Decision Processes

Example: Pacman



- Agent can move up, right, down, and left.
- Moving on a square with a ghost gives a reward (and ghost vanishes).
- The game state changes in each iteration with constant probability α .

Markov Decision Processes

MDP Syntax and Semantics

A **Markov Decision Process** (MDP) is a triple $(S, A, \mathbb{P})$, where

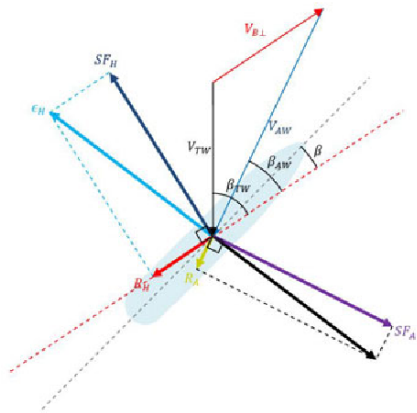
- S is a discrete set of **agent internal states**,
- $A := \bigcup_{s \in S} A_s$, and A_s is a discrete set of possible **actions** in state $s \in S$, and
- $\mathbb{P}(s', r | s, a)$ with $s, s' \in S, a \in A_s, r \in \mathbb{R}$ defines the **transition distribution**.

The agent starts in a (maybe random) **initial state** s_0 and, in an endless loop, in each time step $t = 0, 1, \dots$

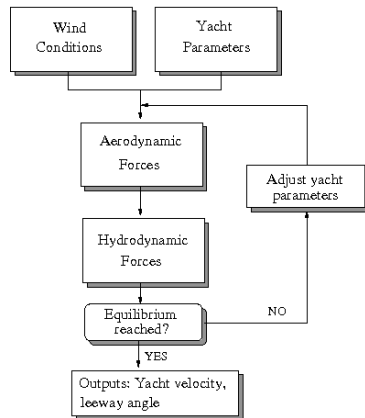
- 1 the agent is told the system state $s_t \in S$ and *must* choose an action in $a_t \in A_{s_t}$,
- 2 the MDP determines a **successor state** and **reward** $(s_{t+1}, r_{t+1}) \sim \mathbb{P}(\cdot | s_t, a_t)$.

Markov Decision Processes

Example: Sailing Domain



(a)



(b)

- Agent can choose an angle to move
- Wind and stream will give random effects to his action

Markov Decision Processes

Absorbing States

How can we model systems that end after some time in an MDP?

Let $S_{\perp} \subseteq S$ be the state of **terminal states**, i.e., states for which the agent cannot perform any action in the real system.

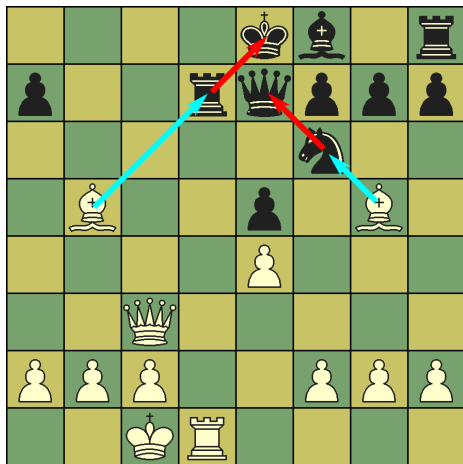
We can use a trick and introduce a synthetic **absorbing** state $s_{\perp}$.

Then, we set for every $s \in S_{\perp} \cup \{s_{\perp}\}$,

- $A_s := \{\perp\}$, i.e., only action $\perp$ is possible
- $\mathbb{P}(s_{\perp}, 0 | s_{\perp}, \perp) = 1$, i.e., no rewards in the absorbing state

Note that there *can* be a reward in the moment when the agent leaves a terminal state (but not after that).

Markov Decision Processes



Given the behavior of an opponent, we also get an MDP for the game against that player.

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Markov Decision Processes

Runs, Trials

A *concrete* sequence of states, actions, and rewards in the MDP is a **run** or **trial**:

$$\tau := s_0, a_0, r_1, s_1, a_1, r_2, s_2, a_2, \dots$$

We distinguish between

- conceptual runs (as above), which are always infinite, and
- observed runs, which are always finite (because game ends or we stop).

Runs in the above examples:

- A full (infinite) Pacman state/action/reward series
- A chess game until check mate or draw followed by $s_{\perp}, \perp, 0\dots$

Course Outline

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Problem Definition

Simplified view on the MDP

We make two simplifying yet usually realistic and not even limiting assumptions:

- ➊ Rewards only depend on the state and not the action or successor state.

We can then say that r_t is the reward when *entering* s_t .

This corresponds to an index shift, and we need a r_0 for s_0 .

- ➋ Rewards are not random but determined by a function $r : S \rightarrow \mathbb{R}$.

We can then simplify the notations of

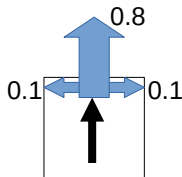
trials to $\tau = s_0, a_0, s_1, a_1, \dots$ and transition probabilities to $\mathbb{P}(s'|s, a)$

Problem Definition

Example

In this section, we consider the following example where all fields have a reward of -0.04 except the field (2,4) and (3,4); the game ends here. The right figure describes $\mathbb{P}$.

	1	2	3	4
3				+1
2				-1
1	Start			



The gray field is a wall, and going there makes the agent being bumped back.

Problem Definition

Overview

An MDP in itself is not a problem yet but only the (agent's view onto the) environment in which decisions *can* be made. The difference MDP vs. environment is crucial!

Problem Statement (intuitive): Find the *best* behavior of an agent in an MDP.

This calls for applying the Bernoulli principle (find utility maximizer) in MDPs, but:

- Over which space should we now define utilities (previously S)?
- What is the search/decision space (previously A)?

Markov Decision Processes

Utility in MDPs

In an MDP, the agent's decisions induce a *run* τ with rewards $r(s_0), r(s_1), \dots$

What is the agent's utility of such a run?

Under rather loose assumptions, it can be shown that the only meaningful utility function is the sum of (discounted) rewards over time produced by (the states of) a run:

$$U(\tau) = \sum_{t=0}^{\infty} \gamma^t r(s_t),$$

where $\gamma \in (0, 1]$ is the **discount factor** for future rewards.

Markov Decision Processes

Policies

In an MDP, the agent want to maximize the expected utility of the *trial*.

But how can the agent control the trial?

Clearly we can look for a good first action a_0 , but what about a_1, a_2 etc. if we don't even know in which states we end up?

Markov Decision Processes

Policies

How would you encode the agent's behavior so that we can create trials?

We encode the behavior of the agent through a **policy**.

A policy π maps each state $s \in S$ to a distribution over actions A_s .

We write $\pi(a|s)$ for the probability that action a is chosen in state s .

We say that π is **deterministic** if, for every $s \in S$, $\pi(a|s) = 1$ for some $a \in A_s$.

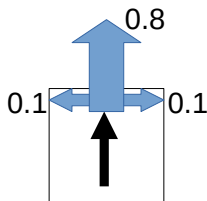
In this case, we write $\pi(s) = a$.

Problem Definition

Example Policy

In our example, a state can be characterized as a field, which allows a convenient way of showing a deterministic policy:

	1	2	3	4
3	→	→	→	+1
2	↑		↑	-1
1	↑	←	←	←



Problem Definition

Random States $\rightarrow$ Random Utility

A run or trial τ is the realization of a **stochastic process**

$$S_0, A_0, S_1, A_1, S_2, A_2, ..$$

The state S_t and action A_t at time t are generally stochastic, even if π is deterministic.

The distribution of state $S_t \sim \mathbb{P}(\cdot | S_{t-1}, A_{t-1})$ depends on the previous state and action.

The distribution of action $A_t \sim \pi(S_t)$ depends on S_t and π .

So even if π is deterministic, we still have that $A_t = \pi(S_t)$ is (generally) stochastic.

Problem Definition

Random States $\rightarrow$ Random Utility

For some $s_0 \in S$, a policy π induces a **Markov chain** $(S_t)_{t \in \mathbb{N}}^{s_0, \pi} := \langle S_0, S_1, \dots \rangle$, in which the distribution of S_t only depends of S_{t-1} , $\mathbb{P}$, and π (except S_0 , which is s_0).

The super-script s_0 indicates that the chain starts in s_0 , and π is used to indicate how the actions are chosen, because this determines the distributions of the S_t .

But thereby also the **utility of a policy**

$$U((S_t)_{t \in \mathbb{N}}^{s_0, \pi}) := U(\langle S_0, S_1, \dots \rangle) = \sum_{t=0}^{\infty} \gamma^t r(S_t)$$

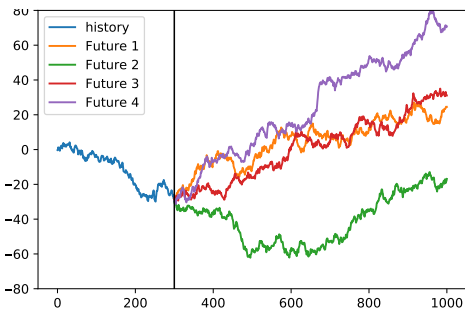
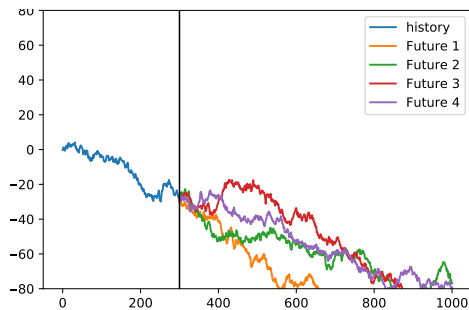
becomes a random variable.

Problem Definition

Example for Stochastic Behavior

The development of our stock portfolio can be seen as a Markov chain.

Every run of the policy induces a distribution of possible futures (colors in figure).



Depending on the policy, these curves can (on average) look quite different.

Problem Definition

Optimal Policies

One reasonable goal is to find a policy π^* such that

$$\pi^* \in \arg \max_{\pi} \mathbb{E}[U((S_t)_{t \in \mathbb{N}})^{s_0, \pi}] = \arg \max_{\pi} \mathbb{E}^{s_0, \pi} \left[\sum_{t=0}^{\infty} \gamma^t r(S_t) \right]$$

i.e., π^* maximizes the **expected utility** across all possible runs.

Here, the annotations of $\mathbb{E}$ have the same meaning as for the Markov chain.

This gives us a formal problem definition.